

Midterm

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Set up tasks

```
# reading in packages
library(tidyverse)
library(janitor)
library(gt)
library(rstatix)
```

```
# reading in halibut data
halibut <- read.csv("halibut.csv")
# reading in bee pathogen data
bee_pathogens <- read.csv("bee-pathogens.csv")
```

Problems

Problem 1. Understanding, critiquing, and adding to written communication

a. Test type

A parametric test that may have been utilized for this analysis is a two-sample t-test, and a nonparametric test that may have been utilized is a Mann-Whitney U-test. A two-sample t-test can be utilized for a comparison of means of two different independent groups. This makes sense for this data set as it is comparing halibut landings (lbs) between Northern and Southern California, where Northern and Southern are the two groups, and the mean of halibut landings (lbs) is used to compare these two groups. Additionally, this test makes sense due the report stating that the commercial landing (lbs) between the two groups are being compared through a p-value, which is done in two sample t-tests. A Mann-Whitney U-test would make sense for this

data set as a non-parametric approach if the data was found to not be approximately normally distributed. A Mann-Whitney U-test compares a central tendency of two independent groups using ranks. This is appropriate for the data set due to it comparing two groups, Northern and Southern California, through a mode of central tendency.

These tests are similar in now they both are comparing a mode of central tendency between two independent groups, result in a probability value, and test a null hypothesis of there being no difference between groups. These tests are different in how they accomplish this comparison. A two-sample t-test utilizes actual values, assumes normality, and compares means. A Mann-Whitney U-test compares the values by ranking them, and therefore does not assume normality. Additionally a Mann-Whitney U-test is comparing medians not means.

b. Writing new statements

Two-sample t-test: Across years (1969-2019), we did not find a significant difference in mean halibut landing weight (lbs) between Northern (mean $\pm$ 95 CI, n = 51) and Southern (mean $\pm$ 95 CI, n = 51) California (Student's two-sample t-test, $t(\text{degrees of freedom}) = \text{test statistic}$, $p = 0.3$, $\alpha = 0.05$).

Mann-Whitney U-test: Across years (1969-2019), we did not find significant difference in median halibut landing weight (lbs) between Northern (median and interquartile range, n = 51) and Southern (median and interquartile range, n = 51) California (Mann-Whitney U-test, $W = \text{test statistic}$, $p = 0.3$, $\alpha = 0.05$).

c. New figures

Figure 1: Two-sample t-test

```
# base layer: ggplot
ggplot(data = halibut,
      # x-axis: location
      # y-axis: landings
      # color by ca_region
      mapping = aes(x = ca_region,
                    y = landings_lb,
                    color = ca_region)) +
# first layer: geom_jitter
geom_jitter(height = 0,
            width = 0.2,
            shape = 21) +
# second layer: error bars to represent 95% CI
stat_summary(fun.data = mean_cl_normal,
```

```

    geom = "errorbar",
    width = 0.1) +
# third layer: points to represent location means
stat_summary(geom = "point",
    fun = mean,
    size = 4) +
# changing theme from the default
theme_classic() +
# relabeling x and y axes
labs(x = "California Region",
    y = "Landings (lbs)") +
# removing legend
theme(legend.position = "none") +
# changing font to serif
theme(text = element_text(family = "serif")) +
# customizing colors
scale_color_manual(
    values = c("Northern" = "palegreen4",
    "Southern" = "pink3"))

```

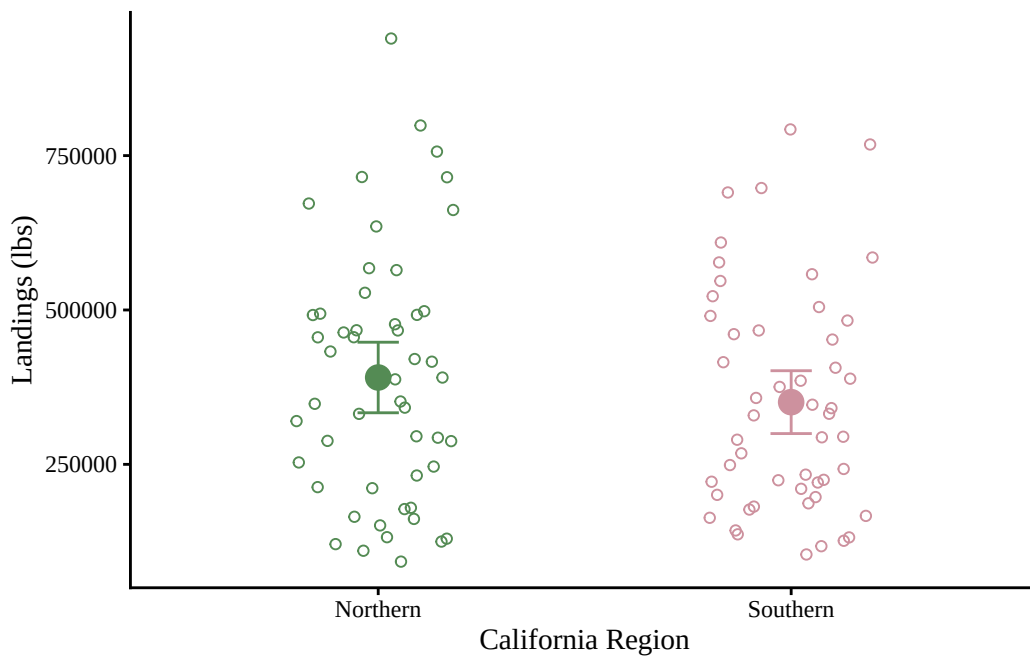
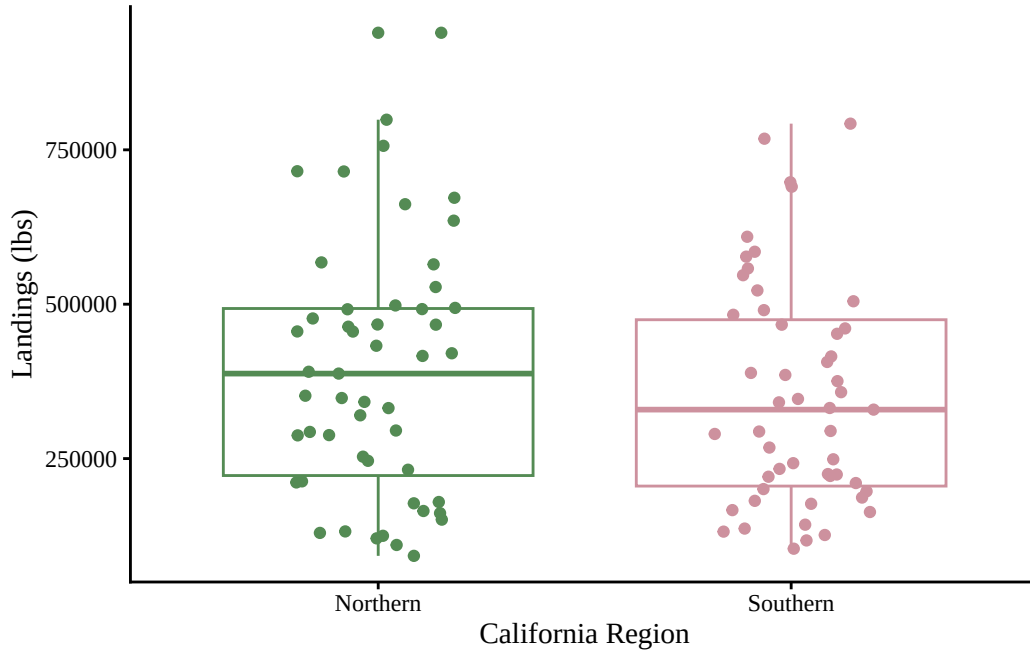


Figure 2: Mann-Whitney U-test

```

# base layer: ggplot
ggplot(data = halibut,
      # x-axis: location
      # y-axis: landings
      # color by ca_region
      mapping = aes(x = ca_region,
                    y = landings_lb,
                    color = ca_region)) +
# first layer: boxplot representing medians
geom_boxplot() +
# second layer: geom_jitter representing individual data points
geom_jitter(height = 0,
            width = 0.2) +
# changing theme from the default
theme_classic() +
# relabeling x and y axes
labs(x = "California Region",
     y = "Landings (lbs)") +
# removing legend
theme(legend.position = "none") +
# changing font to serif
theme(text = element_text(family = "serif")) +
# customizing colors
scale_color_manual(
  values = c("Northern" = "palegreen4",
            "Southern" = "pink3"))

```



d. Caption writing

Figure 1. Commercial California halibut (*Paralichthys californicus*) landings (lbs) in Northern and Southern California from 1969 to 2019. Northern California data are displayed in muted green (palegreen4) and Southern California data is displayed in dusty pink (pink3). Underlying open circle points represent yearly observations of commercial halibut landings. Prominent filled points represent the means and an error bar display $\pm$ 95% confidence intervals for each region. Data source: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Figure 2. Commercial California halibut (*Paralichthys californicus*) landings (lbs) in Northern and Southern California from 1969 to 2019. Northern California data are displayed in muted green (palegreen4) and Southern California data is displayed in dusty pink (pink3). Underlying open circle points represent yearly observations of commercial halibut landings. Boxplots display the median, interquartile range, and overall distribution of landings for each region. Data source: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 2. Data Visualization

a. Plan your figures

- x-axis (name the specific column): “year”
- y-axis (name the specific column): “landings_lb”
- geometry function (hint: there is only one): `geom_line()`
- colors and line types (name the specific column): `color = ca_region`, `linetype = ca_region`
- relabeling the x- and y-axes (name the specific function): `labs()`
- using custom colors (name the specific function): `scale_color_manual()`

theme arguments and the `element_` for:

- positioning the x-axis text at a 45° angle: `theme(axis.text.x = element_text(angle = 45))`
- putting the legend in the top right of the plot: `theme(legend.position = c(0.95, 0.95), legend.justification = c("right", "top"))`
- removing the panel background: `theme(panel.background = element_blank())`
- removing the panel grid: `theme(panel.grid = element_blank())`
- removing the axis ticks: `theme(axis.ticks = element_blank())`
- removing the legend title: `theme(legend.title = element_blank())`

b. Code your figure

Figure 3: Halibut Timeseries

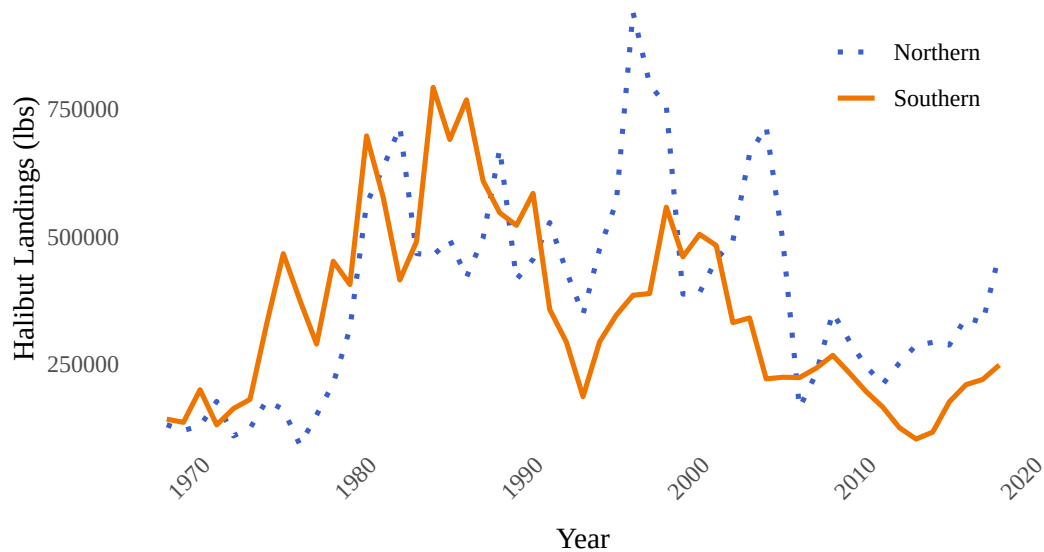
```
# base layer: ggplot
ggplot(data = halibut,
      # x axis year
      # y axis landings_lbs
      # assigning color and line type based on ca_region
      mapping = aes(x = year,
                    y = landings_lb,
                    color = ca_region,
                    linetype = ca_region)) +
# first layer: geom_line
geom_line(linewidth = 0.9) +
# renaming x and y axes
# creating a subtitle of my name
labs(x = "Year",
     y = "Halibut Landings (lbs)",
     subtitle = "Elise LaFreniere") +
# adding custom colors
```

```

scale_color_manual(
  values = c(
    "Northern" = "royalblue3",
    "Southern" = "darkorange2")) +
# assigning line type
scale_linetype_manual(
  values = c(
    "Northern" = "dotted",
    "Southern" = "solid"
  )) +
# adjusting the theme
theme(
  # tilting the x-axis text 45 degrees
  axis.text.x = element_text(angle = 45, hjust = 0.5),
  # moving the legend to the top right
  legend.position = c(0.95, 0.95),
  legend.justification = c("right", "top"),
  # removing the panel background
  panel.background = element_blank(),
  # removing the panel grid
  panel.grid = element_blank(),
  # removing the axis ticks
  axis.ticks = element_blank(),
  # removing the legend title
  legend.title = element_blank(),
  # changing font to be consistent with prior figures
  text = element_text(family = "serif"),
  # moving the subtitle to make the figure look better
  plot.subtitle = element_text(margin = margin(t = 20)))

```

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c. Write a caption

Figure 3. Annual California halibut (*Paralichthys californicus*) landings (lbs) from 1970-2020. Landings are compared between Northern California, shown in the solid green line, and Southern California, shown in the pink dashed line. The data indicates a peak in Northern California landings (lbs) in approximately the late 1990s and a peak in Southern California landings (lbs) in approximately the mid-1980s. Data source: California Department of Fish and Wildlife. 2022. California halibut, *Paralichthys californicus*, Enhanced Status Report. Data accessed 2026-05-04.

Problem 3. Reproducing an analysis

a. Variables

The response variable is the Black Queen Cell Virus loads found in the honey bees. This is a continuous quantitative variable that measures the concentration of viral particles present in an individual bee.

The predictor variable is the habitat type, either being a valley, forest, or edge. This variable indicates the environment in which the honey bee samples were collected.

The researchers measured the concentration of Black Queen Cell Virus load in an individual bee through quantitative PCR. QPCR was done for each pathogen; DMV, BQCV, *Vairimorpha bombi*, and *Crithidia bombi* (“Pathogen Quantification”, paragraph 1).

b. Test type

The researchers utilized a Kruskal-Wallis test to identify a general difference in BQCV loads across the three habitats. This was then followed by a Dunn post-hoc test and Bonferroni correction to identify the specific pairs and adjust the p-value (“Statistical Analysis”, paragraph 1).

c. Hypotheses

H_0 : There is no significant difference in the median Black Queen Cell Virus load across the three habitat types: valley, forest, and edge.

H_A : There is a significant difference in the median Black Queen Cell Virus load across the three habitat types: valley, forest, and edge.

d. Cleaning and organizing data

```
# creating a new object called bee_pathogens_clean
bee_pathogens_clean <- bee_pathogens |>
  # cleaning the names
  clean_names() |>
  # selecting for desired columns
  select(bqcv,
         site_type_x) |>
  # dropping na values
  drop_na(bqcv,
         site_type_x) |>
  # renaming the site_type_x column to be habitat
  rename(habitat = site_type_x) |>
  # capitalizing habitat values
  mutate(habitat = str_to_title(habitat))
# displaying 10 rows using slice_sample
bee_pathogens_clean |>
slice_sample(n = 10)
```

```

      bqcv habitat
1  5.311343   Edge
2  3.037826 Valley
3  6.869602 Valley
4  0.000000 Forest
5  9.708310   Edge
6  0.999518 Forest
7  7.773242   Edge
8  5.090602   Edge
9  5.030311   Edge
10 6.226015 Valley

```

e. Summarizing and table displays

```

# creating new object called bee_pathogens_summary from bee_pathogens_clean
bee_pathogens_summary <- bee_pathogens_clean |>
  # grouping data by habitat
  group_by(habitat) |>
  # using summarize sample to calculate the mean and the sample size
  summarize(
    median_bqcv = median(bqcv, na.rm = TRUE),
    sample_size = n())

```

```

# creating table to display data
bee_pathogens_summary |>
  # using gt() to create table
  gt() |>
  # creating a table header about data
  tab_header(
    title = "Median BQCV Loads by Habitat") |>
  # relabeling the column names
  cols_label(
    habitat = "Habitat Type",
    median_bqcv = "Median BQCV Load",
    sample_size = "Sample Size (n)") |>
  # rounding values to the one decimal point using fmt_number
  fmt_number(
    columns = median_bqcv,
    decimals = 1) |>
  # using tab_source_note to add note about BQCV units
  tab_source_note(

```

```

source_note =
  md("Note: BQCV loads represented as log<sub>10</sub> genome copies")) |>
# changing table font to serif to be consistent with figures
tab_options(
  table.font.names = "serif") |>
# new function
tab_options(latex.tbl.pos = "h",          # h for render the table "here"
            latex.use_longtable = TRUE,    # setting PDF rendering option
            table.width = px(500))        # making the table wider

```

Median BQCV Loads by Habitat

Habitat Type	Median BQCV Load	Sample Size (n)
Edge	6.3	27
Forest	5.0	24
Valley	7.6	29

Note: BQCV loads represented as log₁₀ genome copies

f. Do a test

```

# running a kruskal-wallis test
kruskal.test(
  # bqcv load as a function of habitat
  bqcv ~ habitat,
  # for cleaned data set
  data = bee_pathogens_clean)

```

Kruskal-Wallis rank sum test

data: bqcv by habitat

Kruskal-Wallis chi-squared = 9.8219, df = 2, p-value = 0.007365

g. Do a post-hoc test

```
# running a dunn test for post hoc
dunn_test(
  # bqc as a function of habitat
  bqc ~ habitat,
  # for clean data set
  data = bee_pathogens_clean,
  # adjusting the p value with bonferroni
  p.adjust.method = "bonferroni")
```

```
# A tibble: 3 x 9
  .y.   group1 group2    n1    n2 statistic      p    p.adj p.adj.signif
* <chr> <chr>  <chr>  <int> <int>    <dbl>  <dbl>  <dbl>  <chr>
1 bqc  Edge    Forest    27    24    -1.83 0.0679  0.204    ns
2 bqc  Edge    Valley    27    29     1.31 0.189   0.567    ns
3 bqc  Forest Valley    24    29     3.13 0.00175 0.00525 **
```

h. Calculate an effect size

```
# calculating effect size for kruskal-wallis
kruskal_effsize(
  # bqc as a function of habitat
  bqc ~ habitat,
  # for clean data
  data = bee_pathogens_clean)
```

```
# A tibble: 1 x 5
  .y.      n effsize method magnitude
* <chr> <int>  <dbl> <chr>  <ord>
1 bqc    80   0.102 eta2[H] moderate
```

i. Written communication

We found a **moderate** ($\eta^2 = 0.102$) difference in **median** BQCV loads between valley, forest, and edge habitats (**Kruskal-Wallis**, $\chi^2(2) = 9.82$, **p** = **0.007**, $\alpha = 0.05$). **Median** BQCV loads were significantly higher in the valley (**Mdn** = **7.6**, **n** = **29**) than the forest (**Mdn** = **5.0**, **n** = **24**; **Dunn's post-hoc test: adjusted p-value** = **0.005**), but the edge habitat (**Mdn** = **5.0**, **n** = **29**) did not significantly differ from either (**Dunn's post-hoc test: adjusted p** > **0.05**).